

Database Technologies-Assignment 3

Note : To solve below queries use “hr” database

```
KD2_98846_Raj>--Assignment No - 3
```

```
-> ^C
```

```
KD2_98846_Raj>use hr;
```

```
Database changed
```

```
KD2_98846_Raj>show tables;
```

```
+-----+
| Tables_in_hr |
+-----+
| countries     |
| departments   |
| employees     |
| job_history   |
| jobs          |
| locations     |
| regions       |
+-----+
```

```
7 rows in set (0.06 sec)
```

```
KD2_98846_Raj>SELECT * FROM employee;
```

```
ERROR 1146 (42S02): Table 'hr.employee' doesn't exist
```

```
KD2_98846_Raj>SELECT * FROM employees;
```

```
+-----+-----+-----+-----+-----+-----+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE |
| JOB_ID      | SALARY     | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-----+-----+-----+-----+-----+-----+
| 100         | Steven    | King      | SKING  | 1.515.555.0100 | 2013-06-17 |
| AD_PRES     | 24000.00  | NULL      | NULL   | 90              |
| 101         | Neena     | Yang      | NYANG  | 1.515.555.0101 | 2015-09-21 |
| AD_VP       | 17000.00  | NULL      | 100    | 90              |
| 102         | Lex       | Garcia    | LGARCIA | 1.515.555.0102 | 2011-01-13 |
| AD_VP       | 17000.00  | NULL      | 100    | 90              |
| 103         | Alexander | James     | AJAMES  | 1.590.555.0103 | 2016-01-03 |
| IT_PROG     | 9000.00   | NULL      | 102    | 60              |
| 104         | Bruce     | Miller    | BMILLER | 1.590.555.0104 | 2017-05-21 |
| IT_PROG     | 6000.00   | NULL      | 103    | 60              |
| 105         | David     | Williams  | DWILLIAMS | 1.590.555.0105 | 2015-06-25 |
| IT_PROG     | 4800.00   | NULL      | 103    | 60              |
| 106         | Valli     | Jackson   | VJACKSON | 1.590.555.0106 | 2016-02-05 |
| IT_PROG     | 4800.00   | NULL      | 103    | 60              |
| 107         | Diana     | Nguyen    | DNGUYEN | 1.590.555.0107 | 2017-02-07 |
| IT_PROG     | 4200.00   | NULL      | 103    | 60              |
```

1. Write a query to get unique department ID from employee table.

```
root>select distinct department_id from employees;
```

department_id
NULL
10
20
30
40
50
60
70
80
90
100
110

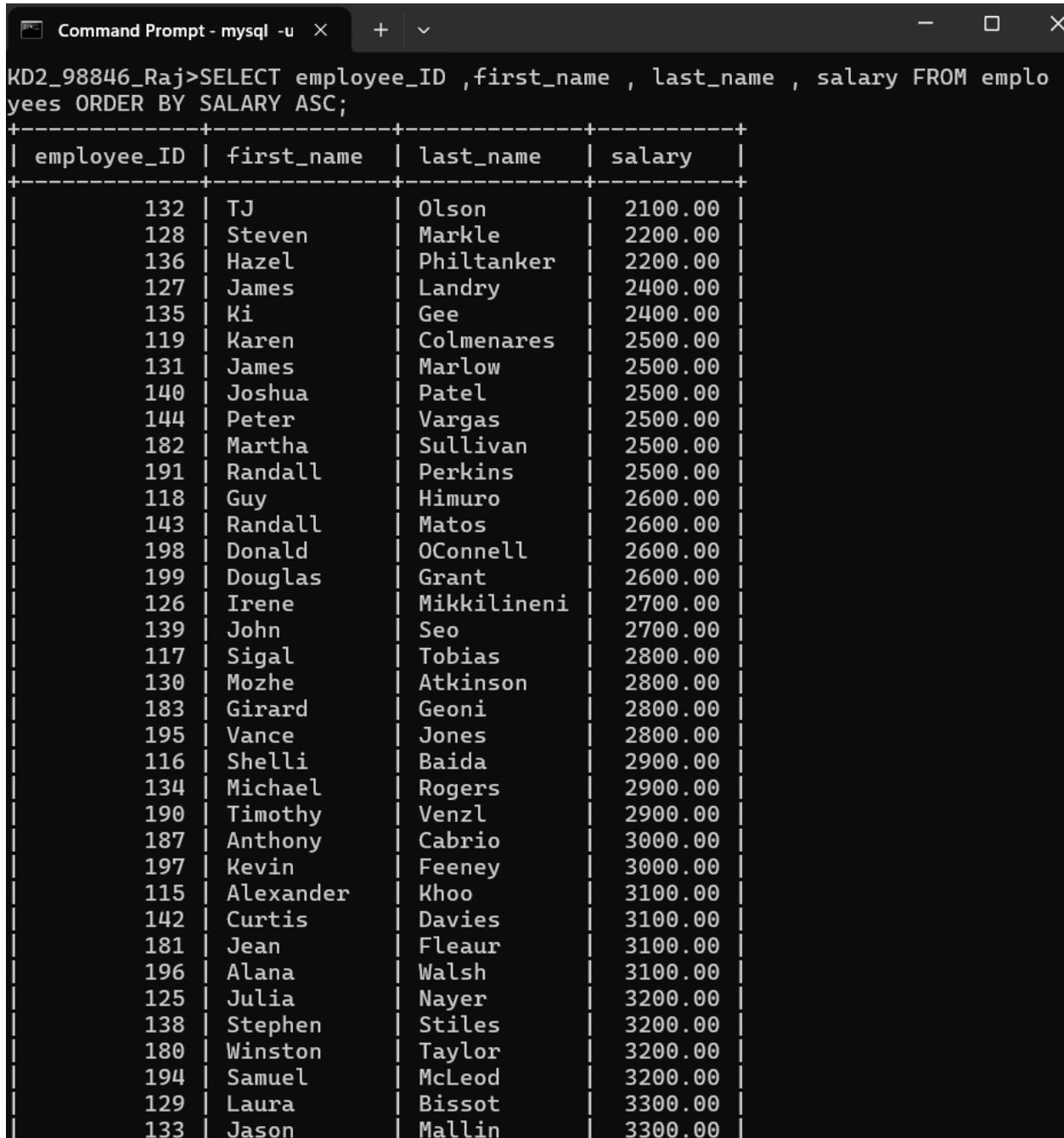
```
12 rows in set (0.06 sec)
```

2. Write a query to get all the employee's details from the employees table displaying the first names in descending order.

```
KD2_98846_Raj>SELECT * FROM employees ORDER BY first_name DESC;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE
JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
180	Winston	Taylor	WTAYLOR	1.650.555.0145	2016-01-24
SH_CLERK	3200.00	NULL	120	50	
171	William	Smith	WSMITH	44.1632.960026	2017-02-23
SA_REP	7400.00	0.15	148	80	
206	William	Gietz	WGIETZ	1.515.555.0171	2012-06-07
AC_ACCOUNT	8300.00	NULL	205	110	
195	Vance	Jones	VJONES	1.650.555.0160	2017-03-17
SH_CLERK	2800.00	NULL	123	50	
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05
IT_PROG	4800.00	NULL	103	60	
141	Trenna	Rajs	TRAJS	1.650.555.0141	2013-10-17
ST_CLERK	3500.00	NULL	124	50	
132	TJ	Olson	TJOLSON	1.650.555.0132	2017-04-10
ST_CLERK	2100.00	NULL	121	50	
190	Timothy	Venzl	TVENZL	1.650.555.0155	2016-07-11
SH_CLERK	2900.00	NULL	122	50	
170	Taylor	Fox	TFOX	44.1632.960025	2016-01-24
SA_REP	9600.00	0.20	148	80	
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07
HR_REP	6500.00	NULL	101	40	
173	Sundita	Kumar	SKUMAR	44.1632.960028	2018-04-21
SA_REP	6100.00	0.10	148	80	

3. Write a query to get the employee ID, names (first_name and last_name), salary in ascending order of salary.



The screenshot shows a Command Prompt window titled "Command Prompt - mysql -u". The user has entered the query: `SELECT employee_ID ,first_name , last_name , salary FROM employees ORDER BY SALARY ASC;`. The results are displayed in a table with four columns: employee_ID, first_name, last_name, and salary. The data is sorted by salary in ascending order.

employee_ID	first_name	last_name	salary
132	TJ	Olson	2100.00
128	Steven	Markle	2200.00
136	Hazel	Philtanker	2200.00
127	James	Landry	2400.00
135	Ki	Gee	2400.00
119	Karen	Colmenares	2500.00
131	James	Marlow	2500.00
140	Joshua	Patel	2500.00
144	Peter	Vargas	2500.00
182	Martha	Sullivan	2500.00
191	Randall	Perkins	2500.00
118	Guy	Himuro	2600.00
143	Randall	Matos	2600.00
198	Donald	OConnell	2600.00
199	Douglas	Grant	2600.00
126	Irene	Mikkilineni	2700.00
139	John	Seo	2700.00
117	Sigal	Tobias	2800.00
130	Mozhe	Atkinson	2800.00
183	Girard	Geoni	2800.00
195	Vance	Jones	2800.00
116	Shelli	Baida	2900.00
134	Michael	Rogers	2900.00
190	Timothy	Venzl	2900.00
187	Anthony	Cabrio	3000.00
197	Kevin	Feeney	3000.00
115	Alexander	Khoo	3100.00
142	Curtis	Davies	3100.00
181	Jean	Fleaur	3100.00
196	Alana	Walsh	3100.00
125	Julia	Nayer	3200.00
138	Stephen	Stiles	3200.00
180	Winston	Taylor	3200.00
194	Samuel	McLeod	3200.00
129	Laura	Bissot	3300.00
133	Jason	Mallin	3300.00

4. Display first name and joining date of the employees who are either IT Programmer or Sales Man.

```
KD2_98846_Raj>SELECT first_name , hire_date FROM employees WHERE job_id = 'IT_PROG' OR job_id = 'sales_man';
```

first_name	hire_date
Alexander	2016-01-03
Bruce	2017-05-21
David	2015-06-25
Valli	2016-02-05
Diana	2017-02-07

5. Display details of employees with employee ID 150 or 160.

```
KD2_98846_Raj>SELECT * FROM employees WHERE employee_id = 150 OR employee_id = 160;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID
150	Sean	Tucker	STUCKER	44.1632.960005	2015-01-30	SA_REP	10000.00	0.30	145
160	Louise	Doran	LDORAN	44.1632.960015	2015-12-15	SA_REP	7500.00	0.30	146

2 rows in set (0.01 sec)

6. Display first name, salary, commission pct, and hire date for employees with salary less than 10000.

```
KD2_98846_Raj>SELECT first_name , salary , commission_pct , hire_date FROM employees WHERE 'salary'<10000;
```

first_name	salary	commission_pct	hire_date
Steven	24000.00	NULL	2013-06-17
Neena	17000.00	NULL	2015-09-21
Lex	17000.00	NULL	2011-01-13
Alexander	9000.00	NULL	2016-01-03
Bruce	6000.00	NULL	2017-05-21
David	4800.00	NULL	2015-06-25
Valli	4800.00	NULL	2016-02-05
Diana	4200.00	NULL	2017-02-07
Nancy	12008.00	NULL	2012-08-17
Daniel	9000.00	NULL	2012-08-16
John	8200.00	NULL	2015-09-28
Ismael	7700.00	NULL	2015-09-30
Jose Manuel	7800.00	NULL	2016-03-07
Luis	6900.00	NULL	2017-12-07
Den	11000.00	NULL	2012-12-07
Alexander	3100.00	NULL	2013-05-18
Shelli	2900.00	NULL	2015-12-24
Sigal	2800.00	NULL	2015-07-24
Guy	2600.00	NULL	2016-11-15
Karen	2500.00	NULL	2017-08-10
Matthew	8000.00	NULL	2014-07-18
Adam	8200.00	NULL	2015-04-10
Payam	7900.00	NULL	2013-05-01
Shanta	6500.00	NULL	2015-10-10
Kevin	5800.00	NULL	2017-11-16
Julia	3200.00	NULL	2015-07-16
Irene	2700.00	NULL	2016-09-28
James	2400.00	NULL	2017-01-14
Steven	2200.00	NULL	2018-03-08

7. Display employees where the first name or last name starts with S.

KD2_98846_Raj>SELECT * FROM employees WHERE first_name LIKE '%S%' OR last_name LIKE '%S%';

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
111	Ismail	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	7700.00	NULL	108	100
116	Shelli	Baida	SBAIDA	1.515.555.0116	2015-12-24	PU_CLERK	2900.00	NULL	114	30
117	Sigal	Tobias	STOBIAS	1.515.555.0117	2015-07-24	PU_CLERK	2800.00	NULL	114	30
123	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	6500.00	NULL	100	50
128	Steven	Markle	SMARKLE	1.650.555.0128	2018-03-08	ST_CLERK	2200.00	NULL	120	50
138	Stephen	Stiles	SSTILES	1.650.555.0138	2015-10-26	ST_CLERK	3200.00	NULL	123	50
139	John	Seo	JSEO	1.650.555.0139	2016-02-12	ST_CLERK	2700.00	NULL	123	50
145	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN	14000.00	0.40	100	80
150	Sean	Tucker	STUCKER	44.1632.960005	2015-01-30	SA_REP	10000.00	0.30	145	80
157	Patrick	Sully	PSULLY	44.1632.960012	2014-03-04	SA_REP	9500.00	0.35	146	80
159	Lindsey	Smith	LSMITH	44.1632.960014	2015-03-10	SA_REP	8000.00	0.30	146	80
161	Sarah	Sewall	SSEWALL	44.1632.960016	2016-11-03	SA_REP	7000.00	0.25	146	80
166	Sundar	Ande	SANDE	44.1632.960021	2018-03-24	SA_REP	6400.00	0.10	147	80
171	William	Smith	WSMITH	44.1632.960026	2017-02-23	SA_REP	7400.00	0.15	148	80
173	Sundita	Kumar	SKUMAR	44.1632.960028	2018-04-21	SA_REP	6100.00	0.10	148	80
182	Martha	Sullivan	MSULLIVA	1.650.555.0147	2017-06-21	SH_CLERK	2500.00	NULL	120	50
184	Nandita	Sarchand	NSARCHAN	1.650.555.0149	2014-01-27	SH_CLERK	4200.00	NULL	121	50
192	Sarah	Bell	SBELL	1.650.555.0157	2014-02-04	SH_CLERK	4000.00	NULL	123	50
194	Samuel	McLeod	SMCLEOD	1.650.555.0159	2016-07-01	SH_CLERK	3200.00	NULL	123	50
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07	HR_REP	6500.00	NULL	101	40
205	Shelley	Higgins	SHIGGINS	1.515.555.0170	2012-06-07	AC_MGR	12000.00	NULL	101	110

22 rows in set (0.01 sec)

KD2_98846_Raj>

8. Display details of jobs in the descending order of the title.

KD2_98846_Raj>SELECT *
-> FROM jobs
-> ORDER BY job_title DESC;

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
ST_MAN	Stock Manager	5500	8500
ST_CLERK	Stock Clerk	2008	5000
SH_CLERK	Shipping Clerk	2500	5500
SA_REP	Sales Representative	6000	12008
SA_MAN	Sales Manager	10000	20080
PU_MAN	Purchasing Manager	8000	15000
PU_CLERK	Purchasing Clerk	2500	5500
PR_REP	Public Relations Representative	4500	10500
AC_ACCOUNT	Public Accountant	4200	9000
IT_PROG	Programmer	4000	10000
AD_PRES	President	20080	40000
MK_REP	Marketing Representative	4000	9000
MK_MAN	Marketing Manager	9000	15000
HR_REP	Human Resources Representative	4000	9000
FI_MGR	Finance Manager	8200	16000
AD_VP	Administration Vice President	15000	30000
AD_ASST	Administration Assistant	3000	6000
AC_MGR	Accounting Manager	8200	16000
FI_ACCOUNT	Accountant	4200	9000

19 rows in set (0.01 sec)

9. Display the details of the employees from department 30, earning the salary greater 10000 but do not receive the commission.

```
KD2_98846_Raj>SELECT *
-> FROM employees
-> WHERE department_id = 30
-> AND salary > 10000
-> AND commission_pct IS NULL;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30

1 row in set (0.01 sec)

10. Display the details of employees reporting to the managers with employee id 100 and 120.

```
KD2_98846_Raj>SELECT *
-> FROM employees
-> WHERE manager_id = 100
-> OR manager_id = 120;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.0
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.0
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.0
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	8000.0
121	Adam	Fripp	AFRIPP	1.650.555.0121	2015-04-10	ST_MAN	8200.0
122	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	7900.0
123	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	6500.0
124	Kevin	Mourgos	KMOURGOS	1.650.555.0124	2017-11-16	ST_MAN	5800.0
145	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN	14000.0
146	Karen	Partners	KPARTNER	44.1632.960001	2015-01-05	SA_MAN	13500.0
147	Alberto	Errazuriz	AERRAZUR	44.1632.960002	2015-03-10	SA_MAN	12000.0

11. Display the unique country_id from locations table

```
KD2_98846_Raj>SELECT DISTINCT country_id
-> FROM locations;
```

country_id
AU
BR
CA
CH
CN
DE
GB
IN
IT
JP
MX
NL
SG
US

14 rows in set (0.04 sec)

12. Display all employees whose have the job_id as IT_PROG and FI_ACCOUNT

```
KD2_98846_Raj>SELECT *
-> FROM employees
-> WHERE job_id = 'IT_PROG'
-> OR job_id = 'FI_ACCOUNT';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALAR
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.
110	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_ACCOUNT	8200.
111	Ismael	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	7700.
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_ACCOUNT	7800.
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	6900.
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.
104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.
105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	4800.
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_PROG	4200.

13. Display all countries in ascending order.

```
KD2_98846_Raj>SELECT *
-> FROM countries
-> ORDER BY country_name ASC;
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	20
AU	Australia	40
BE	Belgium	10
BR	Brazil	20
CA	Canada	20
CN	China	30
DK	Denmark	10
EG	Egypt	50
FR	France	10
DE	Germany	10
IN	India	30
IL	Israel	30
IT	Italy	10
JP	Japan	30
KW	Kuwait	30
ML	Malaysia	30
MX	Mexico	20
NL	Netherlands	10
NG	Nigeria	50
SG	Singapore	30
CH	Switzerland	10
GB	United Kingdom of Great Britain and Nort	10
US	United States of America	20
ZM	Zambia	50
ZW	Zimbabwe	50

25 rows in set (0.02 sec)